

# Data Management and Transformation

**EVR 628- Intro to Environmental Data Science**

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## Exercise 1: Reading and transforming data

Your boss asked you what sounds like a simple query:

How much money have tuna purse seiners made since 2000 when fishing for bigeye tuna (*Thunnus obesus*) in the Eastern Pacific Ocean?

Let's make some assumptions that will help us answer this question:

1. We will interpret “making money” as revenue, not profits
2. The market price of tuna since 2000 has remained relatively stable, at around US\$2/Kg (See Sibert et al. (2012))
3. We will focus on tuna production in the Eastern Pacific Ocean as reported by the IATTC

## Part A) Obtaining data from the wild

How to find the data:

1. Go to [iattc.org](http://iattc.org)
2. In the top menu, **hover** over DATA
3. Click on “Public domain”
4. You will be taken to a page titled “Public domain data for download”
5. We will use “EPO total estimated catch by year, flag, gear, species”
6. Click on `CatchByFlagGear.zip` to the right of the table to prompt a download
7. Save the zip file to inside your EVR628 project at: `data/raw/`<sup>1</sup>
8. Using your finder / explorer, navigate to `EVR628/data/raw/` and unzip the `CatchByFlagGear.zip` file<sup>2</sup>
9. You will get a new folder called `CatchByFlagGear`
10. Read the PDF file enclosed, which contains the documentation

## Part B) Reading data

0. **Put your post-it up**
1. Start a new script called `tuna_analysis` and save it to your `scripts/03_analysis` folder
2. Add a comment header and outline, and load the `tidyverse` package at the top of your script
3. Use the `read_csv()` function to load the new data and assign it to an object called `tuna_data`
4. What are the existing column names?<sup>3</sup>
5. **Remove your post-it when you are done**

Warning: package 'ggplot2' was built under R version 4.5.2

Warning: package 'tidyr' was built under R version 4.5.2

Warning: package 'dplyr' was built under R version 4.5.2

---

<sup>1</sup>If your web browser didn't allow you to specify the download folder, your file is likely in the “Downloads” folder. Navigate there and copy it to the `data/raw/` folder.

<sup>2</sup>Windows users: You might have to click a button called “Extract” in the top of your explorer window.

<sup>3</sup>Hint: use the `colnames()` function

```
# Load packages
library(tidyverse)
library(janitor)

# Load the data
tuna_data <- read_csv("data/raw/CatchByFlagGear/CatchByFlagGear1918-2023.csv")
```

Rows: 13595 Columns: 5

-- Column specification -----

Delimiter: ","

chr (3): BanderaFlag, ArteGear, EspeciesSpecies

dbl (2): AnoYear, t

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# Check colnames after cleaning names
colnames(tuna_data)
```

```
[1] "AnoYear"          "BanderaFlag"      "ArteGear"         "EspeciesSpecies"
[5] "t"
```

## Part C) Renaming columns with `clean_names()` and `rename()`

### 0. Put your post-it up

1. In your console, install the `janitor` package using `install.packages("janitor")`
2. Load `janitor` at the top of your script, and then read the documentation for the `clean_names()` function
3. Modify your code for the `tuna_data` object so that you pipe into `clean_names()` after reading the data
4. What are the new column names?
5. Extend the pipeline above so that we can use the `rename()` function. Rename the columns so that we only retain the English portion of the name. Let's also rename `t` as `catch`
6. **Remove your post-it when you are done**

```
# Load packages
library(tidyverse)
library(janitor)

# Load data
```

```
tuna_data <- read_csv("data/raw/CatchByFlagGear/CatchByFlagGear1918-2023.csv") |>
# Clean column names
clean_names() |>
# Rename some columns
rename(year = ano_year,
       flag = bandera_flag,
       gear = arte_gear,
       species = especies_species,
       catch = t)
```

Rows: 13595 Columns: 5

-- Column specification -----

Delimiter: ","

chr (3): BanderaFlag, ArteGear, EspeciesSpecies

dbl (2): AnoYear, t

i Use `spec()` to retrieve the full column specification for this data.

i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
# Check column names
colnames(tuna_data)
```

```
[1] "year"    "flag"    "gear"    "species" "catch"
```

Congratulations, you now have a tidy data set with which we can work! The next steps are to keep the data we care about, calculate revenues, and then calculate summaries. Let's do that.

## Part D) Filtering rows with filter()

### 0. Put your post-it up

1. What are the unique species represented in the data?
2. What are the fishing gears represented in the data?
3. Does the documentation say what these codes are?<sup>4</sup>
4. What is the species code for bigeye tuna?
5. What is the gear code for purse seine?
6. Create a new object called `ps_tuna_data` that takes the `tuna_data` and filters it to retain data for:

---

<sup>4</sup>Hint: There is a cryptic link to the [reference codes](#)

- a. bigeye tuna
- b. caught by tuna purse seiners
- c. since 2000

## 7. Remove your post-it when you are done

```
# Check unique values for the species column (using a pipe)
tuna_data$species |> unique()
```

```
[1] "SKJ" "YFT" "SWO" "BZX" "ALB" "PBF" "BET" "BLM" "BUM" "MLS" "CGX" "MZZ"
[13] "BKJ" "BIL" "SKH" "TUN" "DOX" "SFA" "SSP" "SRX" "BXQ"
```

```
# Check unique values for the gear column, without a pipe
unique(tuna_data$gear)
```

```
[1] "LP" "PS" "UNK" "HAR" "LTL" "RG" "LL" "GN" "OTR" "LHP" "TX"
```

```
# Create a new data set called ps_tuna_data after filtering
ps_tuna_data <- tuna_data |>
  filter(species == "BET",
         gear == "PS",
         year >= 2000)
```

## Part E) Creating new columns with mutate()

### 0. Put your post-it up

1. Modify the `ps_tuna_data` pipeline to create a new column called `revenue` that calculates the revenue generated by selling the catch<sup>5</sup>
2. Make sure you calculate revenues in Millions of USD
3. Remove your post-it when you are done

```
# Create a new data set called ps_tuna_data after filtering
ps_tuna_data <- tuna_data |>
  filter(species == "BET", # Retain BET values only
         gear == "PS",    # Retain PS values only
         year >= 2000) |> # Retain data from 2000 onwards
  mutate(revenue = catch * 1000 * 2 / 1e6) # Calculate revenue
```

---

<sup>5</sup>Hint: If I catch 10 kilos and the price per kilo is US\$2, then I make US\$20 because  $10 * 2 = 20$

## Part F) Calculating group summaries with `group_by()` and `summarize()`

### 0. Put your post-it up

1. The data right now report catch at the year-by-flag level. Modify the `ps_tuna_data` pipeline so that we have total catch and revenue by year.<sup>6</sup>
2. Remove your post-it when you are done

```
ps_tuna_data <- tuna_data |>
  filter(species == "BET", # Retain BET values only
         gear == "PS",    # Retain PS values only
         year >= 2000) |> # Retain data from 2000 onwards
  mutate(revenue = catch * 1000 * 2 / 1e6) |> # Calculate revenue
  group_by(year) |> # Specify that I am grouping by year
  # Tell summarize that I want to collapse the catch column by summing all its values
  summarize(catch = sum(catch),
            revenue = sum(revenue)) # Same, but for revenues
```

### ! Important

During class we only calculated total revenue. The above code calculates total revenue AND total catch.

## Part G) Visualize the data and answer the question

Remember, the question was:

How much money have tuna purse seiners made since 2000 when fishing for bigeye tuna (*Thunnus obesus*)?

The question is ambiguous because one could answer “They have made X M USD since 2000” or “Every year since 2000, they have made Y M USD per year.” So let’s get both:

### 0. Put your post-it up

1. What is the total revenue?<sup>7</sup>
2. What is the average annual revenue?<sup>8</sup>
3. Build a time-series showing revenues by year. Make sure to correctly label the axis and include a title and caption describing the figure and the data source, respectively.
4. Remove your post-it when you are done.

---

<sup>6</sup>Hint: You will need to use the `group_by`, `summarize()`, and `sum()` functions.

<sup>7</sup>Hint: Use `$` and `sum()`

<sup>8</sup>Hint: Use `$` and `mean()`

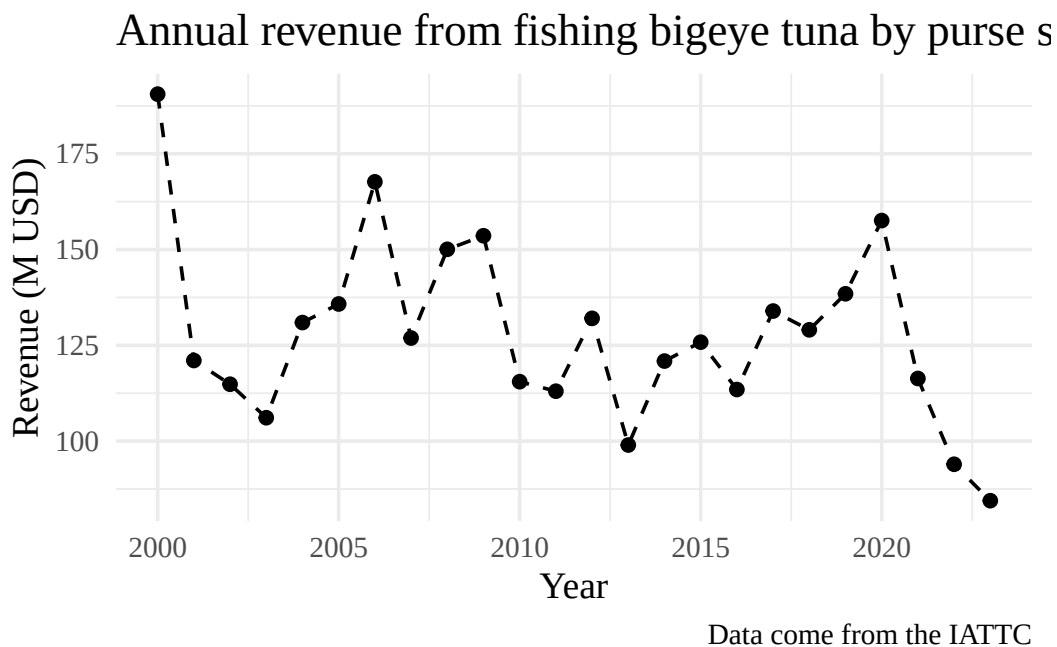
```
# Get total revenue
sum(ps_tuna_data$revenue)
```

```
[1] 3070.97
```

```
# Get mean annual revenue
mean(ps_tuna_data$revenue)
```

```
[1] 127.9571
```

```
# Build plot
ggplot(data = ps_tuna_data,                                # Specify my data
       mapping = aes(x = year, y = revenue)) +             # And my aesthetics
  geom_line(linetype = "dashed") +                         # Add a dashed line
  geom_point() +                                           # With points on top
  labs(x = "Year",                                         # Add some labels
       y = "Revenue (M USD)",
       title = "Annual revenue from fishing bigeye tuna by purse seine vessels",
       caption = "Data come from the IATTC") +
  # Modify the theme
  theme_minimal(base_size = 14,                            # Font size 14
               base_family = "Times")                     # Font family Times
```

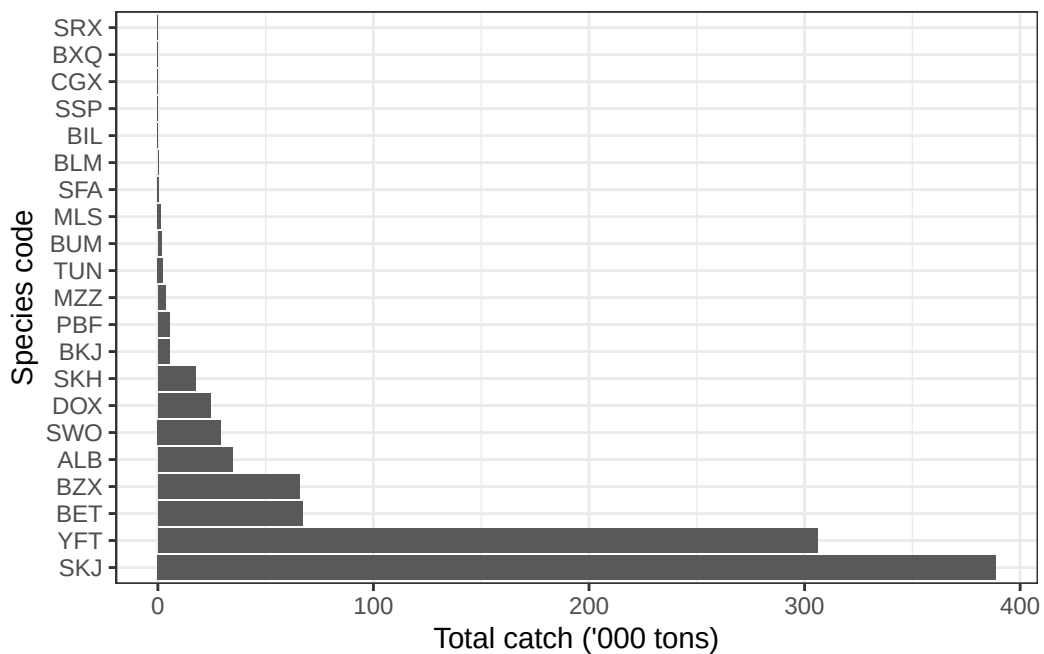


## Extra exercises for you to practice

1. Make a figure showing total catch by species during 2023

```
# Build a new data.frame that has catch by species (in thousand tons)
catch_2023 <- tuna_data |>
  filter(year == 2023) |>
  group_by(species) |>
  summarize(total_catch = sum(catch) / 1e3)

# Now build the figure
ggplot(data = catch_2023,
       aes(x = fct_reorder(species, total_catch, .desc = T), # I am using this fct_reorder t
          y = total_catch)) +
  geom_col() +
  labs(x = "Species code",
       y = "Total catch ('000 tons)") +
  coord_flip() +
  theme_bw()
```



2. Which country caught the most tuna during 2020?



```
# There is more than one way to do this one
# Option 1

# Build a data that has total catch by country in 2020
country_catch <- tuna_data |>
  filter(year == 2020) |> # Retain only data from 2020
  group_by(flag) |> # Get total catch by flag (i.e. sum catch across all species)
  summarize(total_catch = sum(catch))

# And then we can use brackets, dollar signs, and boolean operators to extract the flag
country_catch$flag[country_catch$total_catch == max(country_catch$total_catch)]
```

```
[1] "ECU"
```

```
# Option2: The way I haven't shown you
tuna_data |>
  filter(year == 2020) |>
  group_by(flag) |>
  summarize(total_catch = sum(catch)) |> # Up until here, the pipeline is the same
  arrange(desc(total_catch)) |> # Then I use the arrange function to sort the data in descending order
  head(1) |> # I then retain only the first row, which now _should_ contain the data I want
  pull(flag) # This is a "tidy" version of using a dollar sign to extract a column
```

```
[1] "ECU"
```

3. For each species, identify the year in which catch was at it's maximum

```
# Option 1: With what you already know
tuna_data |>
  group_by(year, species) |>
  summarize(total_catch = sum(catch)) |> # We first calculate total catch by species and year
  group_by(species) |> # Then we group by species
  filter(total_catch == max(total_catch)) |> # And use the filter function. Since the data are grouped by species, this works
  select(species, year_of_max_catch = year) |> # And we keep the columns we care about
  arrange(species)
```

```
`summarise()` has regrouped the output.
i Summaries were computed grouped by year and species.
i Output is grouped by year.
i Use `summarise(.groups = "drop_last")` to silence this message.
i Use `summarise(.by = c(year, species))` for per-operation grouping
  (`?dplyr::dplyr_by`) instead.
```

```
# A tibble: 21 x 2
# Groups:   species [21]
  species year_of_max_catch
  <chr>      <dbl>
1 ALB          2014
2 BET          2000
3 BIL          2013
4 BKJ          2016
5 BLM          1973
6 BUM          1963
7 BXQ          2019
8 BZX          2023
9 CGX          1983
10 DOX         2009
# i 11 more rows
```

```
# Option 2: Using slice_max
tuna_data |>
  group_by(year, species) |>
  summarize(total_catch = sum(catch)) |>
  group_by(species) |>
  slice_max(total_catch) |>
  select(species, year_of_max_catch = year) |>
  arrange(species)
```

`summarise()` has regrouped the output.  
 i Summaries were computed grouped by year and species.  
 i Output is grouped by year.  
 i Use `summarise(.groups = "drop\_last")` to silence this message.  
 i Use `summarise(.by = c(year, species))` for per-operation grouping  
 (`?dplyr::dplyr\_by`) instead.

```
# A tibble: 21 x 2
# Groups:   species [21]
  species year_of_max_catch
  <chr>      <dbl>
1 ALB          2014
2 BET          2000
3 BIL          2013
4 BKJ          2016
5 BLM          1973
6 BUM          1963
```

```

7 BXQ                2019
8 BZX                2023
9 CGX                1983
10 DOX               2009
# i 11 more rows

```

4. How many species have been caught by Mexican-flagged vessels since 2000?

```

# Option 1: A pipeline that ends in a vector, with unique and length
# Start from tuna_data
tuna_data |>
  filter(flag == "MEX") |> # Retain observations associated with Mexico
  pull(species) |> # Pull the species column away from the data.frame, at this point we have
  unique() |> # Get a unique list of species
  length() # Count the number of unique species

```

```
[1] 21
```

```

# Alternatively, retain the data.frame structure
tuna_data |>
  filter(flag == "MEX") |> # Retain observations associated with Mexico
  group_by(flag) |>
  summarize(n_species = n_distinct(species)) # The n_distinct() function is a tidy version of

```

```

# A tibble: 1 x 2
  flag  n_species
<chr>    <int>
1 MEX         21

```

Sibert, John, Inna Senina, Patrick Lehodey, and John Hampton. 2012. "Shifting from Marine Reserves to Maritime Zoning for Conservation of Pacific Bigeye Tuna (*Thunnus Obesus*)."  
*Proc. Natl. Acad. Sci. U. S. A.* 109 (October): 18221–25.